

Align Technology

NASDAQ : \$12.94B mkt cap · 71M sh · \$1.1B cash, zero debt · \$1.10B cash, no debt (net cash, p_fail ~0); ~71.3M shares; ~\$200M buyback active;
ALGN · Revenue \$4.035B FY25 (+1%, flat-lining); GAAP op margin compressed to 13.5% (non-GAAP ~21%); FCF ~\$0.5B; ~70% gross margin;
Mature- ~\$175, YTD ~+7-8%, 52-wk \$122-\$208 - deeply de-rated from the 2021 ~\$700 peak (fwd P/E ~15)
Company DCF

\$181.49

THE CENTRAL QUESTION

Does clear-aligner category growth and Invisalign share retention outrun ASP/share erosion from Angelalign and at-home commoditization?

Align trades at ~\$12.5B (~\$174.84, deeply de-rated from the 2021 ~\$700 peak to ~\$175) - ~25x FCF on \$4.04B revenue growing low-single-digit, ~70% gross margin, net cash and a buyback. It is the category-defining clear-aligner brand (Invisalign + iTero) in a market still only ~15-20% penetrated, but growth has flat-lined (+1%) as discretionary demand softens and Angelalign/at-home rivals pressure ASPs (-1 to -2% guided). The DCF asks whether category penetration plus the teen/Palatal-Expander TAM outrun price/share erosion. The finding lands ~fair (-1.5%): the modal case is +1%, de-rated but not cheap.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$172.27 expected fair value today
-5.1% vs spot \$181.49

FORWARD COMPOUNDED VALUE

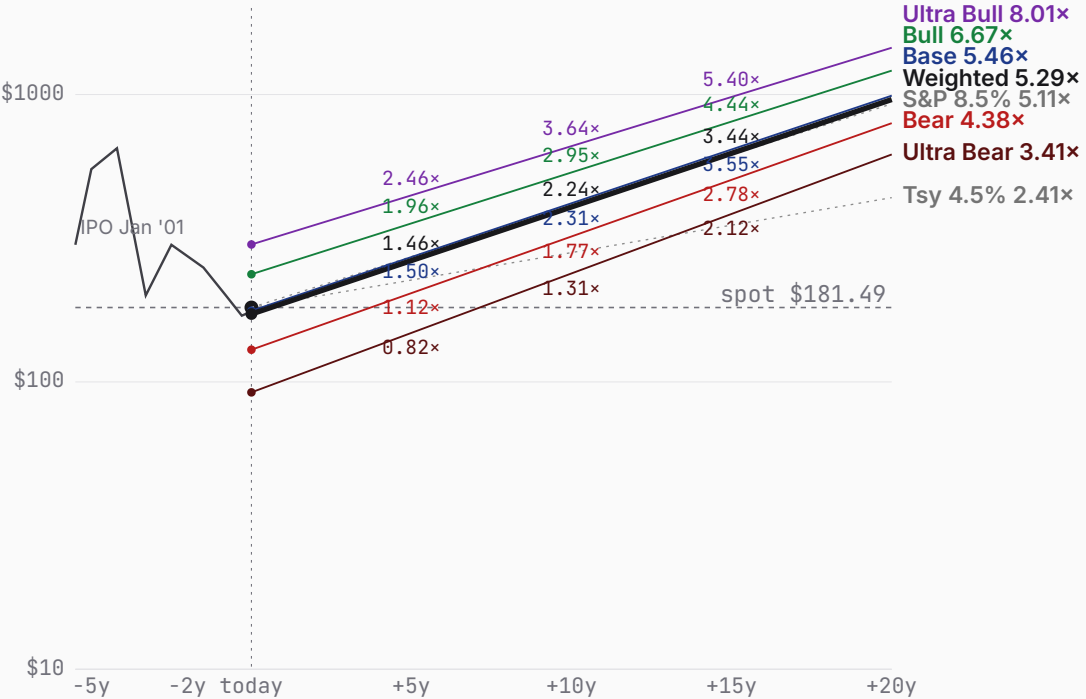
+5y	+10y	+15y	+20y
\$264.49	\$406.29	\$624.48	\$960.41
1.46x spot	2.24x spot	3.44x spot	5.29x spot

WEIGHTING RATIONALE

Ultra Bear 14% Demand stalls; ASP/share erode; ~\$92 (-47%).
Bear 27% Low-single-digit grind; ASP pressure caps it; ~\$129 (-26%).
Base 34% Category penetration resumes; margin firms; ~\$177 (+1%).
Bull 17% Teen/IPE TAM expands; share holds; ~\$237 (+35%).
Ultra Bull 8% Category compounds; Invisalign re-rates; ~\$301 (+72%).

Bull (17%) + Ultra Bull (8%) together contribute \$64.33 — 37% of the \$172.27 expected value despite 25% combined probability. Today's spot (\$181.49) sits 5% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$91.97	BEAR	\$129.43	BASE	\$176.81	BULL	\$236.91	ULTRA BULL	\$300.75
	-49.3% vs spot		-28.7% vs spot		-2.6% vs spot		+30.5% vs spot		+65.7% vs spot
CAGR +1.4% WACC 10.0% CAGR +1.4% Prob 14%		CAGR +3.8% WACC 9.5% CAGR +3.8% Prob 27%		CAGR +6.6% WACC 9.0% CAGR +6.6% Prob 34%		CAGR +8.4% WACC 8.5% CAGR +8.4% Prob 17%		CAGR +10.0% WACC 8.2% CAGR +10.0% Prob 8%	
Demand stalls; ASP/share erode.		Low-single-digit grind; ASP pressure caps it.		Penetration resumes; margin firms.		Teen/IPE TAM expands; share holds.		Category compounds; Invisalign re-rates.	

PROBABILITY WEIGHTS

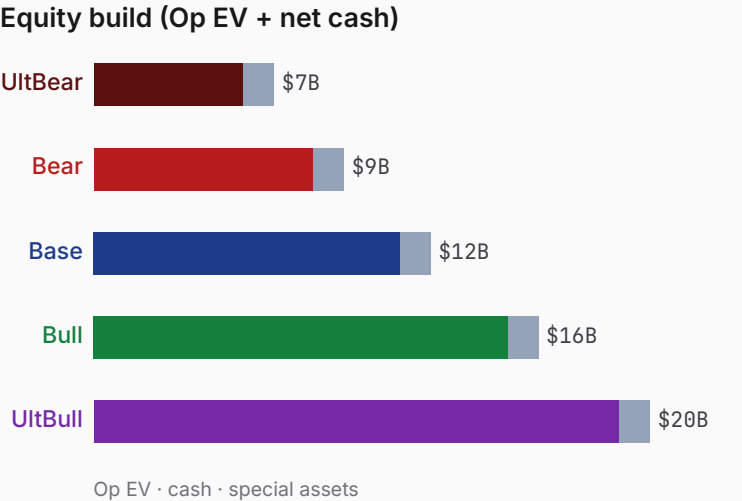
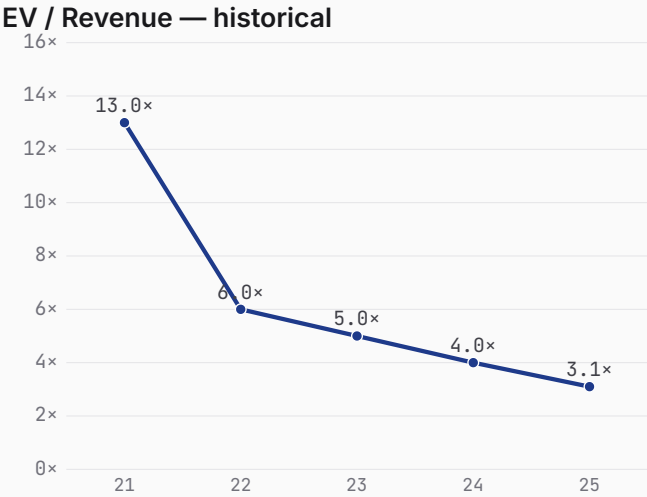
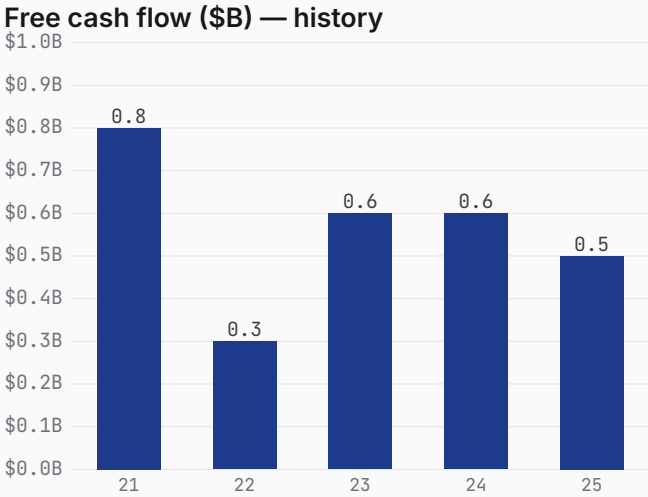
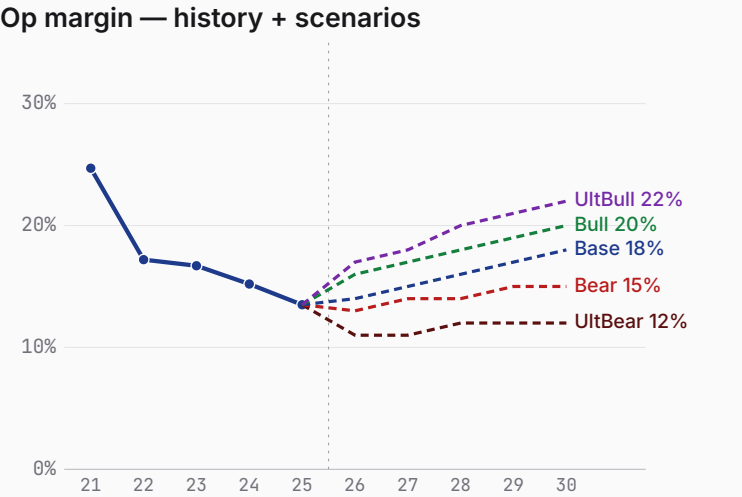
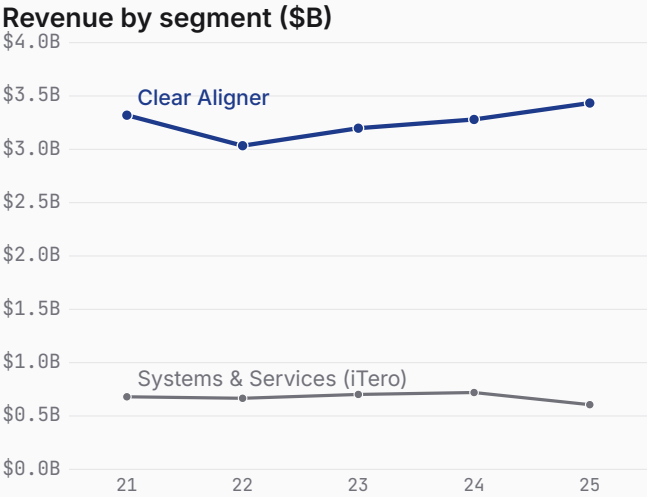
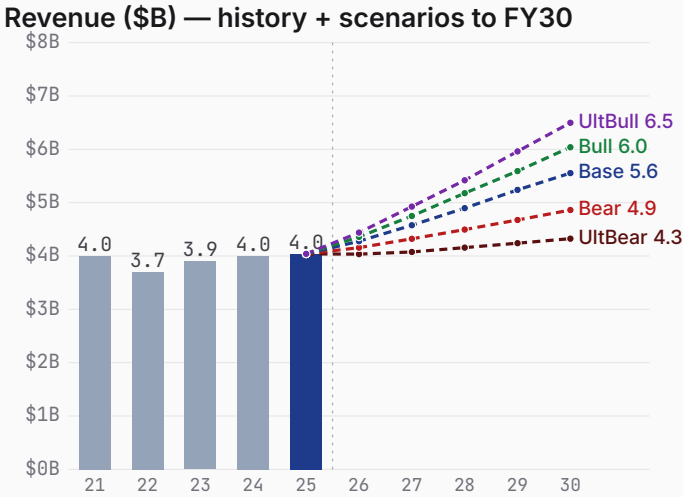
Ultra Bear 14% Bear 27% Base 34% Bull 17% Ultra Bull 8% · Weighted expected \$172.27 (-5.1% vs spot)

ULTRA BEAR	\$91.97	BEAR	\$129.43	BASE	\$176.81	BULL	\$236.91	ULTRA BULL	\$300.75
	-49.3% vs spot		-28.7% vs spot		-2.6% vs spot		+30.5% vs spot		+65.7% vs spot
Probability 14%		Probability 27%		Probability 34%		Probability 17%		Probability 8%	
Demand stalls; ASP/share erode.		Low-single-digit grind; ASP pressure caps it.		Penetration resumes; margin firms.		Teen/IPE TAM expands; share holds.		Category compounds; Invisalign re-rates.	
WHY 14%		WHY 27%		WHY 34%		WHY 17%		WHY 8%	
The genuine bear: revenue is already flat (+1% FY25), op margin has slid four straight years (24.7%→13.5%), ASPs are guided down, and Angelalign is suing on patents while taking China share. 14% weight on price/share erosion beating category growth.		Share retention continues but the discretionary backdrop + ASP pressure + Chinese competition cap the multiple. 27% as the plausible 'holds the lead but doesn't convert to growth' path, consistent with the flat top line and guided ASP declines.		Requires penetration-led case growth + the teen/IPE TAM to offset ASP erosion - consistent with the record Q1 case volume (686,000/+6.7%) and reaffirmed guidance, but not a step-change re-acceleration. 34% as the central, near-fair outcome.		The IPE clearance + record teen case volume + a still-underpenetrated category make a genuine re-acceleration credible. ~17% weight on the TAM expansion landing while share holds.		Everything works - penetration AND the teen/IPE TAM AND share defense AND a margin/multiple re-rate, the high-quality-compounder outcome. ~8%, the long-duration tail.	
WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS	
The discretionary big-ticket consumer stays soft and Angelalign/at-home commoditization wins on price: case volume flat-lines, ASPs keep falling beyond the guided -1 to -2%, and a richer low-cost mix de-levers a ~70% gross margin into low-double-digit operating margins. Revenue is essentially flat; the category-penetration story stalls while share leaks.		Category penetration keeps volume rising but ASP/mix erosion and a competitive market hold growth to ~4% and margins below the historical peak. Invisalign defends its lead without re-accelerating; teen/IPE helps at the edges but doesn't change the trajectory. Revenue compounds ~4% at a low-teens FCF margin.		The modal path: clear-aligner penetration of the ~15-20%-penetrated ortho market resumes mid-single-digit case growth, the teen/Palatal-Expander TAM adds incremental volume, and iTero Lumina anchors the installed base - blending to ~7% revenue growth at a firming ~16-18% operating margin as ASP pressure is offset by volume. Revenue compounds ~7% toward ~\$5.5B.		Clear-aligner category penetration accelerates and the teen/Palatal-Expander launch opens the larger teen ortho TAM (the majority of global case-starts), while Invisalign holds share through patents + the iTero installed base. Volume growth and operating leverage push margins back toward 20%. Revenue compounds ~8-9%.		The full second act: clear aligners take a decisive share of the global ortho market, teen/IPE becomes a durable growth leg, iTero scan-data + AI deepen the cornered resource, and Align re-rates to a high-quality medtech compounder at ~70% gross margin. Revenue compounds ~10% to peak margins.	
A category leader whose volume and price are both pressured earns a low-growth ~13x terminal multiple → DCF ~\$92 (-47%). Net cash sets a floor but can't offset a stalled top line and a margin that keeps grinding lower.		DCF ~\$129 (-26%) - a defended-but-decelerating franchise at ~15x FCF is worth below today's price; the market's discount is warranted if growth stays mid-single-digit and ASPs keep slipping.		DCF ~\$177 (+1%) - essentially at spot. A category-defining brand with a long penetration runway is genuinely valuable, but at ~16x FCF and mid-single-digit growth the modal case only recovers to about today's price; the de-rate corrected the 2021 excess to fair, not to a bargain.		DCF ~\$237 (+35%) - if the penetration + teen-TAM engine re-accelerates and ASPs stabilize, the de-rated entry re-rates with the fundamentals; this is the core of the bull.		DCF ~\$301 (+72%) - the brand reclaims a premium medtech multiple as category growth + share defense + margin recovery all land; ~8% probability, the asymmetric upside the depressed entry is buying.	

Align Technology business snapshot

Bear \$129.43 Base \$176.81 Bull \$236.91

FY21-FY25 history + FY26-FY30 scenario projections · fiscal years end late Dec · Q1 FY2026 (Apr 2026) results



Sources: SEC XBRL company facts (CIK 1097149), Align Q1 FY2026 release (revenue \$1,040M/+6.2%, Clear Aligner \$856M, iTero/Systems \$184M, record 686,000 Invisalign cases/+6.7%, gross margin 70.8%, FY26 ASPs guided -1 to -2%). Revenue is a growth path off FY25; FCF = revenue x FCF margin; Gordon terminal. Multiples vs Dentsply, Straumann, Angelalign.

Align Technology 7 Powers · the moat, scored

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Arena. Clear-aligner orthodontics / dental devices - Align Technology (Invisalign, the category-defining brand, + iTero scanners) vs Angelalign, SmileDirect-style at-home/DIY, Dentsply Sirona, Straumann/ClearCorrect, and traditional wires/brackets; auditing a brand + cornered-resource Power against commoditization and a discretionary-demand market.

POWER AUDIT
dominant-power durability: medium

POWER AUDIT · 7 POWERS

Scale Economies ■■■	Largest clear-aligner manufacturer; manufacturing/R&D/sales leverage at \$4B revenue and ~70% gross margin - real, but Angelalign is scaling fast in China.
Network Economies ■■■	Invisalign-trained doctor base + iTero/exocad workflow create two-sided pull (more doctors → more patients), but not a hard network effect.
Counter-Positioning ■■■	Was the disruptor of wires/brackets; now the incumbent being counter-positioned by lower-cost Angelalign and at-home/DIY models.
Switching Costs ■■■	Doctors invested in iTero hardware + Invisalign training face friction switching systems; patient-side switching is low (a single discretionary purchase).
Branding · thesis leans here ■■■	The dominant Power: Invisalign is the category-defining consumer brand with pricing power and direct-to-consumer demand pull - but eroding at the ASP line under competition.
Cornered Resource ■■■	iTero installed base (>125,000), accumulated scan-data, and a deep patent estate (the basis of the Angelalign suits/ITC action) - durable, the hardest piece to replicate.
Process Power ■■■	Mass-customization manufacturing + the digital treatment-planning stack; a real learning curve, modest as a standalone Power.

COMPETITIVE READ

Align's brand Power is genuine - Invisalign is the category-defining clear-aligner name with pricing power and consumer demand pull, reinforced by the iTero installed base and patent estate as a cornered resource - but the franchise is decelerating: revenue is flat (+1%), op margin has slid four straight years to 13.5%, and ASPs are guided down under Angelalign + at-home price competition. The Audit: branding is durable (medium) but eroding at the price line; the cornered resource (iTero/scan-data/patents) is the real defense, and the live threats are commoditization + a discretionary-demand market. The ~75% de-rate from the 2021 peak corrected the prior excess to roughly fair, not to a bargain; the DCF tests whether category penetration + the teen/IPE TAM outrun ASP/share erosion, and the modal case lands +1% - de-rated, but priced about right.

PEER SET

Angelalign Fast-growing Chinese clear-aligner challenger; competes on price, taking China share - the live patent/ITC threat.	30% gr · 25% mgn · HK-listed
Dentsply Sirona Diversified dental incumbent (incl. SureSmile aligners + imaging); the legacy-dental comp.	0% gr · 10% mgn · ~12x EV/EBITDA
Straumann / ClearCorrect Premium dental/implant group with a growing aligner line; the high-end aligner challenger.	10% gr · 25% mgn · ~30x P/E
At-home / DIY (SmileDirect legacy) Direct-to-consumer aligners at the low end; SmileDirect went bankrupt, but the price-competitive segment persists.	· private

THREAT VECTORS · FALSIFIERS

Branding · Angelalign + at-home/DIY (price competition) falsifier: ASPs fall faster than the guided -1 to -2% as the brand premium erodes against lower-cost aligners.
Cornered Resource · Angelalign (patents / ITC) falsifier: The patent suits + ITC import-block action fail, letting competitors replicate the technology and scan-data edge.
Scale Economies · Discretionary-demand softness + China bulk-procurement falsifier: Case volume flat-lines or declines and a low-cost mix de-levers the ~70% gross margin below ~13% operating margin.

Align Technology

People · Opportunity · Context · Deal

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Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Joseph M. Hogan

11 yrs in seat

1.9%

insider ownership

LOW

key-person risk

Capital allocation (realized)
Joe Hogan (ex-GE/ABB) has been CEO since 2015 — a professional manager; the founders are long gone. The balance sheet is debt-free (~\$1.0B net cash) and FCF-positive (FY2025 revenue \$4.0B, GAAP net income \$410M). Capital return is buyback-only (a \$1B program, ~\$800M remaining), no dividend, with a FY2025 restructuring to lift margins.

Incentive alignment
CEO pay is ~91% at-risk/performance-based. The standout alignment signal is positive: Hogan is a net open-market BUYER — he has bought ~\$8M of stock since 2021 with no sales — uncommon among large-cap CEOs. Say-on-pay drew ~89% support (a notable ~11% dissent on large equity grants). Activist Elliott built a stake (March 2026), no board seats yet.

GOVERNANCE FLAGS

- single class of common stock, one vote per share — no dual-class, no controlling shareholder
- Chair and CEO are SEPARATE, with an independent Chair (transitioning to Kevin Conroy mid-2026)
- ~90% independent board (9-10 of 10-11); declassified annual elections; all committees independent
- notable say-on-pay dissent (~11%) tied to CEO equity-grant size — a compensation watch-item
- an activist (Elliott) is building a stake — external pressure, outcome unresolved

PEOPLE READ

Clean structure plus a rare positive: single-class one-vote, a separate independent Chair, a ~90% independent declassified board, a debt-free balance sheet — and a CEO who buys his own stock in the open market (no sales since 2021). The watch-items are a ~11% say-on-pay dissent on pay size and a fresh Elliott activist stake, neither structural. Key-person risk is low — a professional, no-control CEO with a deep bench.

THE FOUR LEGS

People

observable composite · 1–5

Opportunity

The scenario distribution · the Page-3 business snapshot · the Arthur Indicator

Context

The 7 Powers analysis (Power Audit)

Deal

Open-market purchase = the deal: entry price vs. the scenario distribution (the ~1.5% finding) + position sizing.

Align Technology

supporting analysis · disclaimers · glossary

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PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

The de-rate corrected an excess, it didn't create a bargain.
Down ~75% from the 2021 peak looks cheap, but at ~25x FCF and low-single-digit growth the modal DCF is +1% - growth, not the multiple, is the constraint.
- 2

The category is still only ~15-20% penetrated.
Clear aligners are early in displacing wires/brackets; teen/IPE opens the larger teen ortho TAM the flat top line masks.
- 3

Brand + installed base are the moat.
Invisalign is category-defining; iTero (>125,000 scanners) + scan-data + the patent estate are hardest for Angelalign to replicate.
- 4

ASP/share erosion is the whole question.
ASPs guided -1 to -2%, Angelalign taking China share + litigating patents, at-home at the low end - 'category grows' vs 'Align grows' is the fork.
- 5

Net cash, p_fail ~0.
\$1.10B cash, no debt, ~\$200M buyback - no solvency risk; the downside is growth/multiple, not survival.

FALSIFICATION TRIGGERS

Bull validation

Invisalign case volume re-accelerates above ~+7% · teen/IPE case-starts scale materially · ASPs stabilize (not down) · operating margin recovers back toward ~20%

Bear validation

case volume flat-lines or declines for 2+ quarters · ASPs fall faster than the guided -1 to -2% · Angelalign/at-home take visible share · op margin grinds below ~13%

Reframe needed

if clear aligners commoditize (Angelalign wins on price + the ITC/patent actions fail), the branding + cornered-resource Powers erode - re-rate the terminal toward a low-growth, price-taking dental-device multiple

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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Clear Aligner — Align's largest segment (~82-85% of revenue) - Invisalign removable aligners that displace traditional wires/brackets; case volume x ASP is the swing factor.

iTero / Systems & Services — Intraoral scanners (iTero Lumina, installed base >125,000) + software/services - the cornered-resource base that feeds Invisalign workflow and scan-data.

ASP — Average selling price per case; guided -1 to -2% in FY26 on country/product mix and competition - the price half of the volume-vs-price question.

IPE (Invisalign Palatal Expander) — US-FDA-cleared device targeting teen/kids case-starts (the majority of global ortho) - the TAM-expansion lever in the bull.

FCF margin — Free cash flow / revenue (~12-13% FY25); the mature-DCF lever - whether category growth converts the ~70% gross margin to free cash flow as ASP pressure bites.